American Computer Science League

2022 Finals ● Short Solutions ● Intermediate/Classroom Divisions

1. Boolean Algebra

$$\overline{A\,B\ +\ (\overline{(B\ +\ C)}\,A)}\ =\ \overline{AB}\ \overline{(B\ +\ C)}A$$
$$=\ (\overline{A}\ +\ \overline{B})\ (A\,\overline{B}\,\overline{C})$$
$$=\ (\overline{A}\ +\ \overline{B})\ (\overline{A}\ +\ B\ +\ C)$$
$$=\ \overline{A}\ +\ \overline{A}\,B\ +\ \overline{A}\,C\ +\ \overline{A}\,\overline{B}\ +\ 0\ +\ \overline{B}\,C$$
$$=\ \overline{A}\ +\ \overline{B}\,C$$

B. $\overline{A}\ +\ \overline{B}\,C$

2. Boolean Algebra

$$(A\ \oplus\ B)\ (\overline{A\,B}\ +\ B) = (A\ \oplus\ B)\ (\overline{A}\ +\ \overline{B}\ +\ B)$$
$$=\ A\ \oplus\ B$$

$\oplus$ is TRUE when values are different. Therefore (0,1) and (1,0) make the expression TRUE. So 2 make it TRUE.

C. 2

3. Bit-String Flicking

(RSHIFT-2 (NOT (RCIRC-6 (LSHIFT-1 (LCIRC-12 10110110)))))
= (RSHIFT-2 (NOT (RCIRC-6 (LSHIFT-1 01101011))))
= (RSHIFT-2 (NOT (RCIRC-6 11010110)))
= (RSHIFT-2 (NOT 01011011))
= (RSHIFT-2 10100100)
= 00101001

D. 00101001

4. Bit-String Flicking

Let X = abcde
LHS = (RSHIFT-1 00111) OR (LCIRC-2 01110) AND (NOT 10101)
= 00011 OR 11001 AND 01010
= 00011 OR 01000
= 01011
RHS = (LCIRC-1 (RCIRC-2 X))
= (LCIRC-1 (RCIRC-2 abcde))
= (LCIRC-1 deabc)
= eabcd
LHS = RHS ⇒ 01011 = eabcd
So a = 1, b = 0, c = 1, d = 1, e = 0 ⇒ 10110

E. None of the above

5. Recursive Functions

$f(17) = 2 * f(14) + 1 = 2 * (-23) + 1 = -45$
$f(14) = 2 * f(11) + 1 = 2 * (-12) + 1 = -23$
$f(11) = f(5) - 3 = -9 - 3 = -12$
$f(5) = f(2) - 3 = -6 - 3 = -9$
$f(2) = f(1) - 3 = -3 - 3 = = -6$
$f(1) = 1 - 4 = -3$

B. -45

6. Recursive Functions

For mod 4 the sequence is:

0, 1, 1, 2, 3, 1, 0, 1, 1, 2, 3, 1, …

Therefore, the Pisano period for mod 4 is 6.

B. 6

7. Digital Electronics

The direct translation of the digital circuit is: $(\overline{A\,B})(\overline{B + C}) + C$
This simplifies as follows:
$(\overline{A\,B})(\overline{B + C}) + C = (\overline{A} + \overline{B})(\overline{B}\,\overline{C}) + C$
$= \overline{A}\,\overline{B}\,\overline{C} + \overline{B}\,\overline{C} + C$
$= \overline{B}\,\overline{C} + C$

A. $\overline{B}\,\overline{C} + C$

8. Digital Electronics

The circuit is: $A \oplus B$ $gate?$ $\overline{B + C}$

Gates: AND - 1 makes it T (1,0,0)
OR - 5 make it T (0,0,0), (0,1,0), (0,1,1), (1,0,0), (1,0,1)
NAND - 7 make it T
NOR - 3 make it T

A. AND

9. Prefix-Infix-Postfix

+ - / + 2 8 5 * 3 + / 9 3 4 ^ 2 4 = + - / (+ 2 8) 5 * 3 + (/ 9 3) 4(^ 2 4)
= + - (/ 10 5) * 3 (+ 3 4) 16
= + - 2 (* 3 7) 16
= + (- 2 21) 16
= + (-19) 16
= -3

A. -3

10. Prefix-Infix-Postfix

2 3 * 4 * 6 / 3 2 * 2 ^ 2 4 ^ 2 + / -
= (2 3 *) 4 * 6 / (3 2 *) 2 ^ (2 4 ^) 2 + / -
= ((* 2 3) 4 *) 6 /((* 3 2) 2 ^) ((^ 2 4) 2 +) /) -
= ((* * 2 3 4) 6 /)((^ * 3 2 2) (+ ^ 2 4 2) /) -
= (/ * * 2 3 4 6) (/ ^ * 3 2 2 + ^ 2 4 2) -
= - / * * 2 3 4 6 / ^ * 3 2 2 + ^ 2 4 2

C. - / * * 2 3 4 6 / ^ * 3 2 2 + ^ 2 4 2

11. Computer Number Systems

$2022_{10} = 3746_8$
The next 3 consecutive octal digits would be: 4012
$4012_8 = 2058_{10}$

D. 2058

12. Computer Number Systems

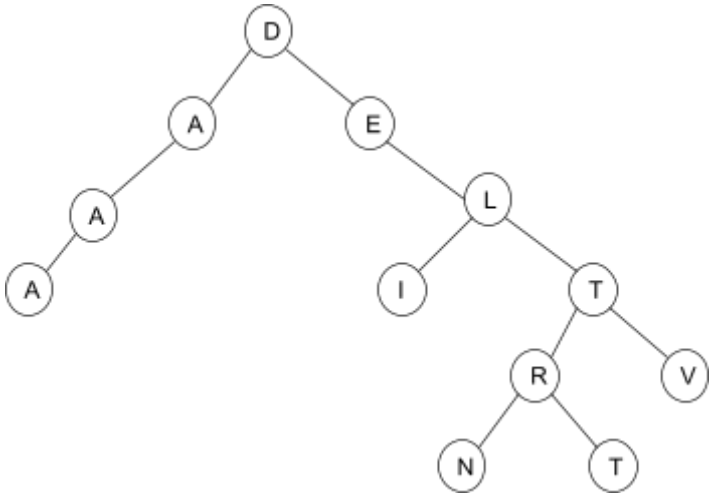
Converting to decimal:
$10110_2 * A_{16} / 13_8 - 20_{10} + (6_8)^2 = 22 * 10 / 11 - 20 + 36$
$= 220 / 11 - 20 + 36$
$= 20 - 20 + 36$
$= 36$

$36_{10} = 24_{16}$

B. 24

13. Data Structures

The binary search tree for **DELTAVARIANT is:**



The internal path length is: 2 * 1 + 2 * 2 + 3 * 3 + 2 * 4 + 2 * 5
= 2 + 4 + 9 + 8 + 10 = 33

C. 33

14. Data Structures

The STACK is constructed using LIFO as follows:

G, EMPTY, I, IN, ING, IN, I, IE, IER, IERB, IERBR, IERB, IER, IERE, IEREA, IEREAD, IEREA, IERE, IER, IE, I

The next item popped is I.

D. I

15. Graph Theory

$$\begin{vmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{vmatrix}^{2} = \begin{vmatrix} 3 & 0 & 2 & 2 & 0 \\ 0 & 1 & 1 & 2 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & 1 \\ 1 & 1 & 2 & 1 & 1 \end{vmatrix}$$

There are 7 paths of length 2 from A (3 0 2 2 0).

C. 7

16. Graph Theory

There are 5 cycles from A: ABA, AFEA, ADCA, AFDCA, and ABDCA.

B. 5

17. What Does This Program Do?

This program checks to see if a number is a factor of a Fibonacci number. Every number is a factor. Therefore there are 10 outputs.

C. 10

18. LISP

(CADDR M) = (CADDR '((a b (d)) (b (c a) d) ((a b) (b) d))))
= (CADR '((b (c a) d) ((a b) (b) d)))
= (CAR '(((a b) (b) d)))
= ((a b) (b) d)

A. ((a b) (b) d)

19. FSAs and Regular Expressions

[^abiou] [hro] [a-j]* (e|s)*

Only 2 items satisfy the regular expression: cheese and coffee

bread - fails at b
fish - fails at i
meat - fails at e
chips - fails at p
chocolate - fails at o
cream - fails at m
pickles - fails at i
grapes - fails at p
cookies - fails at the 2nd o
crackers - fails at k

B. 2

20. Assembly Language

The assembly program translates to the following BASIC program:

```
    t = 0
    n3 = 8
    n2 = 5
10  n1 = n3 - n2
    t = n3 * n3 + t
    n3 = n2
    n2 = n1
    if n1 > 0 then 10
    t = t + 1
    print t
```

This produces 8^2 + 5^2 + 3^2 + 2^2 + 1^2 + 1 = 104 (also 8 * 13). In fact, the sum of the squares of the first N Fibonacci numbers equals the product of the largest number and the next one ((N+1)th) in the sequence.

C. 104